University of Niagara Falls, Canada

Capstone Project Proposal

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Title of Capstone Project

Analyzing Profit Erosion from Product Returns in E-Commerce: A Multi-Method Analytics Framework

Project Idea

Product returns present a significant challenge for e-commerce organizations, not only operationally but economically. Prior research has shown that returns can materially affect customer profitability and firm-level financial performance (Petersen & Kumar, 2009). While much existing analytics work focuses on predicting return likelihood or improving reverse logistics efficiency, this project reframes returns as **economic reversal events** that directly erode realized revenue and margin.

By combining transaction-level data with customer-level behavioral patterns, the study aims to develop a comprehensive analytical framework that quantifies profit erosion attributable to returns, examines customer behavioral patterns through segmentation, explores predictive approaches for identifying high-risk customers, and analyzes key factors associated with profit erosion magnitude. This multi-method approach: integrating descriptive analysis, unsupervised learning, predictive modeling, and econometric regression—will illustrate what information may be relevant, how an analytics pipeline can be applied, and propose a framework that is intended to be repeatable across similar contexts for **understanding the magnitude and drivers of profit erosion from product returns** in the e-commerce industry. Optionally, if time permits, the study will explore prescriptive analytics to identify optimal intervention thresholds.

Problem Identification

Product returns are commonly monitored using operational metrics such as return counts, service-level indicators, or customer satisfaction scores. However, these measures often fail to capture the **economic consequences** of returns. When a product is returned, firms experience an immediate reversal of realized revenue and gross margin, followed by additional costs related to

customer service handling, inspection, unpacking, shelving, and restocking—activities that have been documented as significant cost drivers in reverse logistics operations (Rogers & Tibben-Lembke, 2001).

Although returns and reverse logistics have been widely studied, prior research has noted that economic value recovery and profitability impacts receive less attention than operational efficiency (Guide & Van Wassenhove, 2009; Stevenson & Rieck, 2024). Additionally, much of the analytics literature emphasizes return rates or volume forecasting rather than **margin reversal and cumulative profit erosion** (Toktay, 2003). Recent work in predictive analytics for returns has demonstrated the potential for machine learning approaches to identify patterns in return behavior (Cui et al., 2020), yet these primarily focus on forecasting volumes rather than quantifying economic impact or identifying actionable intervention strategies.

This gap limits an organization's ability to identify which products and customers drive the greatest financial exposure from returns to segment customers based on behavioral risk profiles, and to develop data-driven intervention strategies that maximize profit protection while minimizing customer friction.

Aim of Investigation

The primary aim of this project is to develop a **repeatable multi-method analytical framework** for quantifying and predicting profit erosion from product returns and demonstrate its application using simulated e-commerce data, where profit erosion is defined as observed margin reversal at the item level, augmented by modeled return-processing costs.

Building on evidence that returns influence customer profitability and firm performance (Petersen & Kumar, 2009), the study employs a comprehensive approach combining descriptive analysis, unsupervised learning, predictive modeling, and econometric regression to:

1. **Quantify** profit erosion differences across product categories and brands
2. **Segment** customers into behaviorally distinct clusters with differential profit erosion profiles

3. **Predict** high profit erosion risk using transaction and behavioral features

4. **Estimate** marginal associations between key behavioral variables and profit erosion magnitude while controlling confounding factors

5. **[Optional]** Explore prescriptive analytics to identify optimal customer intervention thresholds

The framework integrates observed margin reversal with modeled return-processing costs, with directional validation against industry benchmarks to assess real-world applicability. The primary contribution is not merely confirming that returns cause profit erosion—this is established—but rather developing a comprehensive, operationalizable analytics pipeline that transforms raw e-commerce transaction data into actionable insights for profit protection.

Research Questions and Hypotheses

Research Question 1 (RQ1) - Descriptive Foundation

Do returned items exhibit statistically significant differences in **profit erosion** across product categories and brands?

- **$H0_1$ (Null):** Mean profit erosion associated with returned items is equal across product categories and brands.
- **$H1_1$ (Alternative):** Mean profit erosion associated with returned items differs significantly across product categories and/or brands.

Justification:
Product categories differ in cost structures, pricing strategies, and handling requirements. Prior research highlights that economic recovery from returns varies across product types (Guide & Van Wassenhove, 2009), suggesting that profit erosion may not be uniform across categories. This foundational analysis informs subsequent modeling by identifying high-risk product segments.

Research Question 2 (RQ2) - Customer Segmentation

Can unsupervised learning identify distinct customer behavioral segments, and do these segments differ significantly in profit erosion intensity?

- **$H0_2$ (Null):** Customer segments identified through clustering algorithms do not differ significantly in mean profit erosion from returns.
- **$H1_2$ (Alternative):** Customer segments identified through clustering algorithms exhibit statistically significant differences in mean profit erosion from returns.

Justification:
Traditional demographic segmentation may not capture the behavioral patterns that drive return-related profit erosion. Unsupervised learning enables data-driven identification of customer typologies based on purchasing and return behaviors without pre-defined constraints (Cui et al., 2020), allowing discovery of high-impact customer groups that may not align with demographic categories.

Research Question 3 (RQ3) - Predictive Modeling

Can machine learning models accurately predict high profit erosion customers using transaction-level and behavioral features, and which features contribute most significantly to prediction accuracy?

- **$H0_3$ (Null):** Machine learning models cannot predict high profit erosion customers with acceptable accuracy (AUC $\leq$ 0.70)
- **$H1_3$ (Alternative):** Machine learning models can predict high profit erosion customers with acceptable accuracy (AUC > 0.70).

Justification:
Predictive modeling enables proactive identification of high-risk customers before substantial profit erosion occurs (Cui et al., 2020). By comparing multiple algorithms and analyzing feature importance, this analysis will reveal which customer and transaction attributes most strongly predict economic exposure, if there are any supporting development of targeted intervention strategies.

Research Question 4 (RQ4) - Econometric Analysis

What are the marginal associations between key behavioral variables (return frequency, basket size, purchase recency) and profit erosion magnitude, controlling for product attributes and customer demographics?

- **$H0_4$ (Null):** Behavioral variables exhibit no statistically significant marginal associations with profit erosion when controlling for product attributes and demographics.
- **$H1_4$ (Alternative):** Behavioral variables exhibit statistically significant marginal associations with profit erosion when controlling for product attributes and demographics.

Justification:

While RQ3 identifies predictive patterns, econometric regression quantifies the conditional associations between specific behaviors and profit erosion magnitude while controlling for confounding factors (Petersen & Kumar, 2009). This enables calculation of marginal associations, answering questions such as "how much does profit erosion increase for each additional return?" which directly informs intervention cost-benefit analysis. Predictive performance will be interpreted in conjunction with precision–recall trade-offs and operational relevance rather than statistical significance alone

Research Question 5 (RQ5) - Prescriptive Analytics [OPTIONAL]

What customer targeting threshold may improve profit protection while balancing intervention costs and customer friction?

- **$H0_5$ (Null):** Intervention strategies based on predictive model outputs do not yield net positive economic value compared to no intervention.
- **$H1_5$ (Alternative):** Intervention strategies based on predictive model outputs yield statistically significant net positive economic value.

Justification:

Predictive models enable risk identification, but prescriptive analysis determines optimal action thresholds by balancing intervention costs against prevented profit erosion (Cui et al., 2020). This transforms analytical insights into actionable business strategies with quantifiable ROI.

Note: This research question will be pursued if time permits after completion of RQ1-RQ4.

Proposed Data Sources

The primary dataset for this project is the thelook_ecommerce public dataset hosted on Google BigQuery (https://console.cloud.google.com/marketplace/product/bigquery-public-data/thelook-ecommerce?project=gen-lang-client-0779042528). Although synthetic, the dataset is well maintained and provides the necessary structure for transaction-level economic analysis, including sale price, product cost, explicit item-level return indicators, product attributes, and customer-level information. Because direct reverse logistics (RL) costs are not fully observable in public datasets, this study operationalizes profit erosion using observed margin reversal and modeled processing cost assumptions. The use of a publicly maintained dataset aligns with recommendations for methodological transparency while enabling end-to-end analytics development (Guide & Van Wassenhove, 2009).

To mitigate the limitations associated with synthetic data, findings from RQ1 **will be directionally validated** using anonymized, aggregated industry benchmarks by comparing category-level return patterns and profit erosion ratios, using data from a U.S.-based educational supplies retailer, School Specialty LLC. This validation approach aligns with established frameworks (Sargent, 2013), focusing on whether the framework reveals patterns consistent with real-world data rather than claiming parameter transferability. RQ2-RQ5 primarily demonstrates the methodological framework using synthetic data, with the design intended for application to proprietary datasets where full validation would be possible.

Intended Outcome

The intended outcome is a **comprehensive profit-centric analytical framework** for e-commerce returns, providing anlytical evidence through:

1. **Descriptive insights** identifying product categories and brands with highest profit erosion
2. **Customer segmentation** revealing behaviorally distinct groups with differential economic impact through unsupervised learning

3. **Predictive capability** to identify high-risk customers with validated accuracy metrics and feature importance analysis
4. **Econometric evidence** quantifying marginal associations between specific behaviors and profit erosion magnitude while controlling confounds
5. **[Optional] Prescriptive recommendations** for optimal intervention thresholds with ROI analysis

By integrating descriptive, predictive, and econometric approaches into a repeatable analytics pipeline, the framework **seeks to address** gaps identified in prior research regarding the economic consequences of returns (Petersen & Kumar, 2009; Rogers & Tibben-Lembke, 2001; Cui et al., 2020; Stevenson & Rieck, 2024). The multi-method approach **aims to demonstrate** the full spectrum of data analytics capabilities—from understanding what exists, to predicting what will happen, to quantifying behavioral associations, to optimizing what actions to take.

If successful, stakeholders including operations managers, marketing teams, and financial analysts **could** use findings to inform return policy design, customer intervention strategies, and product portfolio decisions. The framework **is designed to be repeatable**, enabling potential application across different e-commerce contexts and product categories, **with directional validation from real-world data intended to assess** its practical utility beyond academic analysis.

References

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